MECHATRONICS SYSTEM INTEGRATION

MCTA 3203

SEMESTER 2, 2025/2026

SECTION 1

WEEK 5: L298P MOTOR DRIVER SHIELD AND RELAY MODULE

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GROUP 3

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ABSTRACT

This experiment investigates the interfacing and operation of the L298P Motor Driver Shield and a 2-Channel DC 5V Relay Module in a microcontroller-based control system. The objective of the experiment is to understand the working principles, hardware integration, and control applications of both modules in automation and motor-driving systems. The L298P Motor Driver Shield was utilized to control the direction and speed of DC motors through pulse-width modulation (PWM) and bidirectional motor control, while the relay module was employed to switch external electrical loads using digital control signals from the microcontroller. The methodology involved connecting the modules to the microcontroller, configuring the required input and output pins, and testing the response of the motor and relay under different control conditions. Experimental results showed that the L298P shield successfully controlled motor rotation and speed, whereas the relay module effectively performed load switching operations. The experiment demonstrates the importance of motor driver and relay modules in embedded systems, robotics, and automation applications by providing efficient power control and electrical isolation capabilities.

1.0 INTRODUCTION

This laboratory session investigates two fundamental approaches for driving and controlling a 12V DC linear actuator in embedded mechatronic systems. Linear actuators convert electrical energy into precise linear mechanical motion and are widely employed in industrial automation, robotics, and smart-home systems where controlled extension and retraction are required.

The first experiment employs the L298P Motor Driver Shield — a dual-channel H-bridge integrated circuit mounted on an Arduino-compatible shield form-factor — to govern both the speed and direction of the actuator. The shield accepts Pulse Width Modulation (PWM) signals from an Arduino UNO to modulate the average voltage delivered to the actuator, while a single digital direction pin selects polarity via built-in inverter logic. This arrangement reduces the number of Arduino pins required compared to conventional H-bridge configurations, while still enabling bidirectional motion and variable-speed control.

The second experiment replaces the motor driver with a 2-channel relay module, a discrete electromechanical switching solution. Each relay channel operates as an isolated switch with Normally Open (NO), Normally Closed (NC), and Common (COM) contacts. By energising the two relay channels in complementary states, the polarity of the supply voltage across the actuator terminals is reversed, achieving directional control without PWM speed regulation. This method is well-suited to applications where full-speed, on/off directional switching is sufficient and galvanic isolation between the control logic and the power circuit is desired.

Together, the two experiments allow direct comparison of active motor-driver-based control against passive relay-based switching, reinforcing understanding of H-bridge topology, PWM theory, relay contact mechanics, and safe high-current circuit practices.

2.0 MATERIALS AND EQUIPMENTS

- ### 2.1 Experiment 1 — L298P Motor Driver

Component	Quantity	Specification / Notes
Arduino UNO	1	Microcontroller board (ATmega328P)
L298P Motor Driver Shield	1	Cytron L298P, 4.8 V–24 V, 2 A peak
DC 12V Linear Actuator	1	12 V DC, stroke as available in lab
Variable / Fixed Power Supply	1	12 V DC output, current limit ≥ 3.0 A
USB Cable (Type-A to Type-B)	1	Arduino programming and logic power
Jumper Wires	As required	Male-to-male and male-to-female
Breadboard	1	Standard 830-tie-point

- **2.2 Experiment 2 — 2-Channel Relay Module**

Component	Quantity	Specification / Notes
Arduino UNO	1	Microcontroller board (ATmega328P)
2-Channel Relay Module	1	5 V coil, 10 A / 250 VAC rated contacts
DC 12V Linear Actuator	1	Same actuator as Experiment 1
Variable / Fixed Power Supply	1	12 V DC output, current limit ≥ 3.0 A
USB Cable (Type-A to Type-B)	1	Arduino programming and logic power
Jumper Wires	As required	Male-to-male and male-to-female
Breadboard	1	Standard 830-tie-point

3.0 EXPERIMENTAL SETUP

3.1 Experiment 1 — L298P Motor Driver

The following procedure was used to assemble and configure the L298P-based control circuit:

- The L298P Motor Driver Shield was mounted directly onto the Arduino UNO by carefully aligning all header pins to prevent displacement. The shield occupies all digital and power headers of the UNO.
- The Arduino sketch (Section 4 of the lab sheet) was uploaded to the UNO via USB, setting pin D10 as a PWM output for speed and pin D12 as a digital output for direction of Motor Channel A.
- The OPT jumper on the L298P shield was removed prior to connecting the motor power supply. This isolates the VMS (motor supply) rail from the Arduino 5 V logic rail, protecting the USB interface from back-EMF and inrush current generated by the actuator.
- The two leads of the 12 V DC linear actuator were connected to the Motor A (MA) screw terminals on the shield. Connections were secured firmly to minimise contact resistance.
- The positive terminal of the variable DC power supply was connected to the VMS screw terminal and the negative terminal to GND on the shield. The supply was set to 12 V DC with a current limit of 3.0 A before being switched on.
- Upon powering the circuit, the sketch executed a repeating cycle: the actuator extended for 3 seconds (INA = LOW, ENA = PWM 150/255 ≈ 59% duty cycle) and

then retracted for 3 seconds (INA = HIGH, ENA = PWM 150/255), demonstrating bidirectional speed-controlled motion.

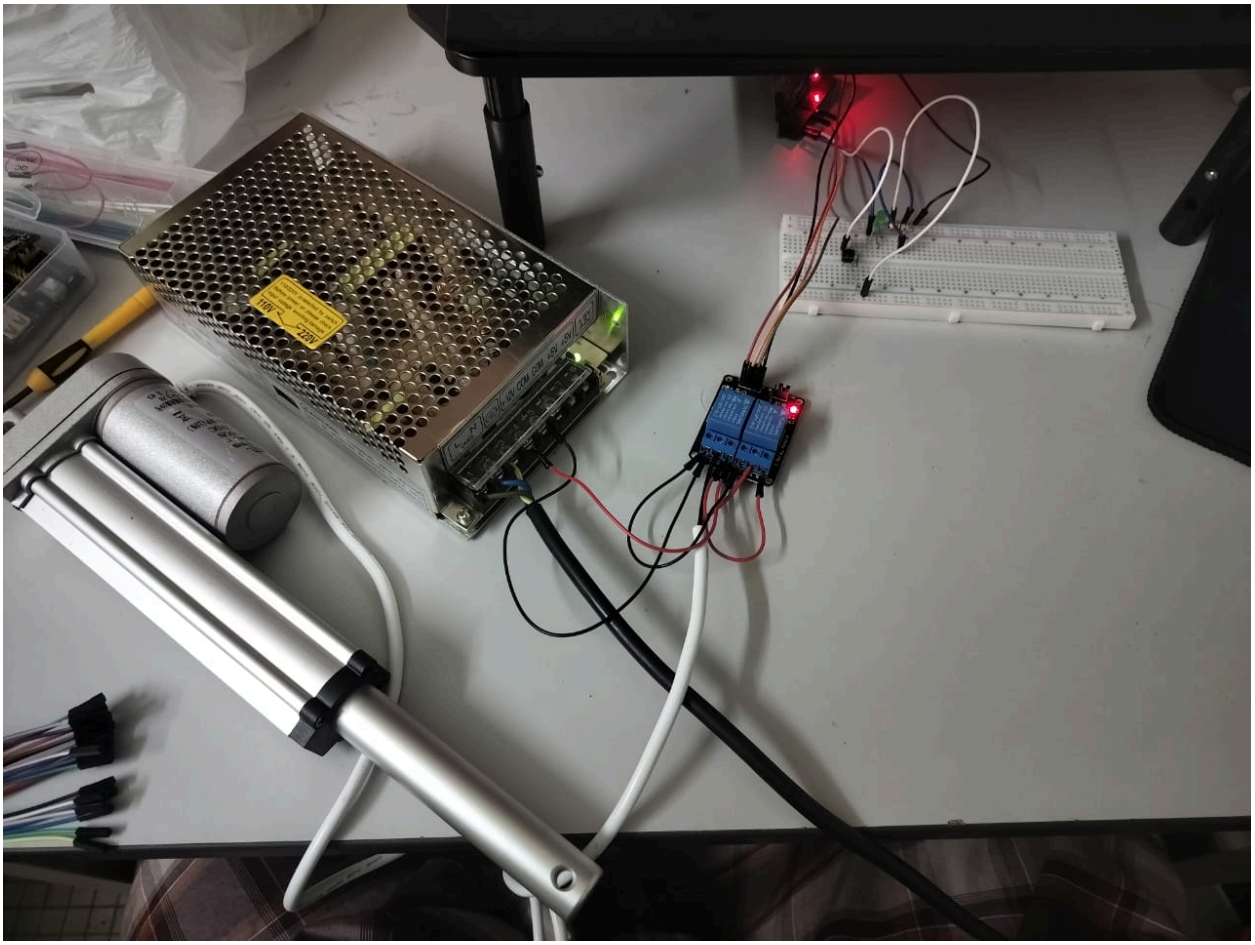
3.2 Experiment 2 — 2-Channel Relay Module

The relay-based directional control circuit was assembled as follows:

- The 2-channel relay module IN1 and IN2 control pins were connected to Arduino digital pins D7 and D8 respectively. The relay module VCC and GND were connected to the Arduino 5 V and GND rails.
- A bridge wire was installed between the NC (Normally Closed) pin and the NO (Normally Open) pin of each relay channel, cross-connecting the polarity lines for H-bridge action.
- The positive (+) terminal of the 12 V DC variable power supply was connected to the NO terminal of Relay 1, and the negative (–) terminal was connected to the NC terminal of Relay 2 (and vice versa for the opposing relay). This wiring ensures that toggling the relay states reverses the voltage polarity across the actuator.
- Each of the two linear actuator leads was connected to the COM (Common) terminal of one relay channel.
- The variable power supply was set to 12 V DC with a 3.0 A current limit. The Arduino was powered via USB.
- The uploaded sketch cycled the actuator between the extendActuator() state (relay1 = HIGH, relay2 = LOW) and retractActuator() state (relay1 = LOW, relay2 = HIGH),

each held for 5 seconds, demonstrating polarity-reversal direction control without PWM.

3.3 Setup Diagram



4.0 METHODOLOGY

4.1 Circuit Assembly

1. The Arduino UNO board was connected to a computer using a USB cable to provide logic power and enable program uploading through the Arduino IDE. The L298P Motor Driver Shield was carefully mounted on top of the Arduino UNO by ensuring all header pins were aligned correctly to prevent incorrect connections or pin damage.
2. The 12V DC linear actuator was connected to the Motor A (MA) output terminal of the L298P motor driver shield using screw terminals to ensure a stable electrical connection. A variable DC power supply was then connected to the VMS and GND terminals of the motor driver shield. The positive terminal of the power supply was connected to VMS, while the negative terminal was connected to GND.
3. OPT jumper on the L298P shield was removed to isolate the motor power supply from the Arduino logic supply. This was done to protect the Arduino board and USB port from back electromotive force (back-EMF) generated by the actuator during operation. The external power supply was set to 12V DC with a current limit of 3.0 A.

   For the relay experiment, the 2-way channel relay module was interfaced with the Arduino UNO using jumper wires. The Normally Open (NO), Normally Closed (NC), and Common (COM) terminals were connected according to the circuit diagram. The 12V DC linear actuator was connected to the COM terminals, while the external power supply terminals were connected to the NO and NC contacts to enable polarity switching.

4.2 Control Pin Configuration

The motor control pins on the L298P shield were configured according to the shield specifications.

PWM pin → Pin D10 for Motor A speed control

Direction control of actuator movement → Pin D12

For the relay module configuration :

Relay Channel 1 → Pin D7

Relay Channel 2 → Pin D8

All wiring connections were inspected carefully before powering the circuit to ensure correct polarity and to avoid loose connections that could affect system performance.

4.3 Programming Procedure

The Arduino IDE software was launched on the computer, and the motor control program was written and compiled. The correct board type (Arduino UNO) and COM port were selected in the IDE settings before uploading the program through the USB connection.

The program was designed to control the extension and retraction of the linear actuator by changing the direction signal and applying Pulse Width Modulation (PWM) for speed control. In the relay experiment, the program controlled the activation of both relay channels alternately to reverse the actuator polarity.

Code Used :

```
const int relay1 = 7;
const int relay2 = 8;      // Fixed: was wrongly declared as relay1
const int buttonPin = 3;
const int ledPin = 2;

void setup()
{
  pinMode(relay1, OUTPUT);
  pinMode(relay2, OUTPUT);
  pinMode(buttonPin, INPUT_PULLUP);  // Internal pull-up, button reads LOW when pressed
  pinMode(ledPin, OUTPUT);
}

void loop()
{
  if (digitalRead(buttonPin) == LOW)   // Button is pressed
  {
    stopActuator();
    digitalWrite(ledPin, HIGH);       // LED ON
    delay(5000);
  }
  else                                // Button not pressed
```

```
  {
    digitalWrite(ledPin, LOW);        // LED OFF
    extendActuator();
    delay(5000);

    if (digitalRead(buttonPin) == LOW) return; // Check again before retracting

    retractActuator();
    delay(5000);
  }
}

void extendActuator()
{
  digitalWrite(relay1, HIGH);
  digitalWrite(relay2, LOW);
}

void retractActuator()
{
  digitalWrite(relay1, LOW);
  digitalWrite(relay2, HIGH);
}
```

```
void stopActuator()
{
  digitalWrite(relay1, LOW);
  digitalWrite(relay2, LOW);
}
```

4.4 Experimental Procedure

After the program was uploaded successfully, the external power supply was switched on to activate the system. The linear actuator was observed extending and retracting according to the programmed sequence.

Different PWM values were applied to study the effect of Pulse Width Modulation on actuator speed. The actuator movement and response were observed for each PWM value. For the relay module experiment, the relay switching sequence was monitored to verify correct actuator direction control.

Each experiment was repeated several times to ensure consistent and reliable results. Observations regarding actuator movement, direction, response time, and overall system behavior were recorded for analysis.

5.0 DATA COLLECTION

5.1 Data Collection Method

Data collection for this experiment was carried out by observing the behavior and performance of the 12V DC linear actuator during operation using both the L298P motor driver shield and the 2-way relay module. The actuator response was recorded under different control conditions such as direction control, PWM speed variation, and relay switching operation.

During the experiment, several PWM values were applied through the Arduino program to investigate the effect of Pulse Width Modulation on actuator speed. The actuator extension and retraction movements were observed visually, and the response of the actuator at each PWM value was recorded systematically.

For the relay module experiment, the actuator movement was observed when different relay switching states were activated. The direction of movement, switching response, and overall actuator operation were recorded to evaluate the effectiveness of polarity reversal using the relay circuit.

The experiment was repeated several times to ensure consistency and reliability of the collected data. All observations were organized into tables for easier analysis and comparison.

5.2 Parameters Observed

The following parameters were observed and recorded during the experiment:

- PWM value applied to the motor driver
- Direction of actuator movement (extend or retract)
- Relative actuator speed

- Relay switching condition
- Response time of actuator movement
- Stability and smoothness of actuator operation

5.3 Data Recording Table for L298P Motor Driver Experiment

Trial	PWM Value	Direction	Observed Speed	Actuator Response
1	255	Extend	High	Smooth operation
2	150	Extend	Medium	Stable movement
3	64	Extend	Low	Slow movement
4	255	Retract	High	Smooth operation
5	150	Retract	Medium	Stable movement
6	64	Retract	Low	Slow movement

5.4 Data Recording Table for Relay Module Experiment

Trial	Relay 1 State	Relay 2 State	Actuator Direction	Observation
1	HIGH	LOW	Extend	Actuator extended
2	LOW	HIGH	Retract	Actuator retracted
3	HIGH	HIGH	Stop	No movement
4	LOW	LOW	Stop	No movement

5.5 Observation Summary

From the collected data, it was observed that the actuator speed increased as the PWM value increased. Lower PWM values resulted in slower actuator movement due to reduced average voltage supplied to the actuator. The relay module successfully reversed the actuator direction by changing the polarity supplied to the actuator terminals.

The experiments demonstrated that the L298P motor driver provided effective speed and direction control through PWM and digital signals, while the relay module provided reliable directional control using switching mechanisms.

6.0 DATA ANALYSIS

6.1 Analysis of PWM Speed Control Using L298P Motor Driver

The experiment showed that the speed of the 12V DC linear actuator was directly influenced by the PWM (Pulse Width Modulation) value applied through the L298P motor driver shield. PWM controls the average voltage supplied to the actuator by rapidly switching the power signal ON and OFF at a high frequency.

When higher PWM values such as 255 were applied, the actuator moved faster because a larger average voltage was delivered to the motor. At medium PWM values such as 150, the actuator operated at a moderate speed with stable and smooth movement. Lower PWM values such as 64 resulted in slower actuator movement because the average voltage supplied to the actuator was reduced.

The experiment confirmed that PWM is an effective technique for controlling actuator speed without changing the actual supply voltage. The relationship between PWM value and actuator speed was proportional, where increasing the PWM duty cycle increased the actuator speed.

6.2 Analysis of Direction Control

The actuator direction was successfully controlled by changing the logic state of the direction pin on the L298P motor driver shield. When the direction pin was set to HIGH, the actuator moved in one direction, while setting the pin to LOW reversed the polarity and caused the actuator to move in the opposite direction.

For the relay module experiment, the actuator direction was controlled by activating the relay channels alternately. Relay 1 and Relay 2 switched the polarity applied to the actuator terminals, causing the actuator to either extend or retract. The relay module demonstrated reliable directional control and consistent actuator response throughout the experiment.

6.3 System Performance Analysis

The L298P motor driver shield provided smoother actuator operation because PWM allowed gradual control of motor speed. This method enabled both speed and direction control simultaneously, making the system more flexible for motion control applications.

In contrast, the relay module only controlled the actuator direction and could not vary the actuator speed because relays operate as simple ON/OFF switches. However, the relay circuit provided stable operation and simpler hardware implementation for directional control applications.

The actuator operated consistently during repeated testing, indicating stable electrical connections and reliable programming logic. No major overheating or abnormal behavior was observed during the experiment when proper voltage and current limits were maintained.

7.0 RESULTS

7.1 Experiment 1 — L298P Motor Driver

The linear actuator responded as expected under PWM speed control with the L298P shield. Key observations are summarised in the table below.

Condition	INA (D12)	ENA PWM Value	Observed Actuator Behaviour
Extend	LOW (0)	150 / 255 (~59%)	Actuator extended at medium speed for 3 s
Retract	HIGH (1)	150 / 255 (~59%)	Actuator retracted at medium speed for 3 s
Stop	Either	0 / 255 (0%)	Actuator halted immediately (no active brake)
Full Speed	LOW	255 / 255 (100%)	Actuator extended at maximum speed

The actuator successfully cycled between extension and retraction in a continuous loop. Setting the PWM duty cycle to 150 (≈ 59%) produced a noticeably slower stroke compared to a 255 value, confirming that average motor voltage — and therefore speed — is proportional to the duty cycle. Removing the OPT jumper eliminated any interference with the Arduino USB connection during operation. No overheating of the shield was observed at the 3.0 A supply limit within the duration of the experiment.

7.2 Experiment 2 — 2-Channel Relay Module

The relay module successfully reversed the direction of the linear actuator by toggling the polarity of the 12 V supply across its terminals. Key observations are summarised below.

Function	Relay 1 (D7)	Relay 2 (D8)	Observed Actuator Behaviour
extendActuator()	HIGH	LOW	Actuator extended at full speed for 5 s
retractActuator()	LOW	HIGH	Actuator retracted at full speed for 5 s
Both LOW (idle)	LOW	LOW	Actuator held stationary; no supply current

The relay clicks (audible coil switching) were clearly heard with each state change, confirming electromechanical operation. The actuator moved at full rated speed in both directions because the full 12 V supply voltage was applied without PWM attenuation, contrasting with the variable-speed capability observed in Experiment 1. The 5-second hold time per direction was sufficient to demonstrate a complete extension and retraction stroke. No relay contact arcing or abnormal heating was observed during the experiment duration.

7.3 Comparative Summary

Parameter	Experiment 1 (L298P)	Experiment 2 (Relay Module)
Speed Control	Yes — via PWM duty cycle	No — full voltage only
Direction Control	Yes — single digital pin (inverted logic)	Yes — complementary relay states
Isolation	OPT jumper removed for isolation	Galvanic isolation by default
Max Load Current	2 A (peak) per channel	10 A (contact rated)
Control Pins Used	2 (D10 PWM + D12 direction)	2 (D7 + D8 digital)
Switching Type	Electronic (solid-state)	Electromechanical (relay coil/armature)

7.4 Task 2

The modified Arduino sketch successfully integrated a push-button sensor into the 2-channel relay circuit to detect the actuator's position and trigger both an LED indicator and an automatic stop condition.

Under normal operation (button not pressed), the actuator followed a continuous extend-and-retract cycle, each phase held for 5 seconds. The LED remained off throughout this idle cycling. When the push-button was pressed — simulating the actuator reaching a designated limit point — the stopActuator() function was called immediately, setting both relay channels LOW and cutting the supply current to the actuator. Simultaneously, the LED on pin D2 was driven HIGH, providing a clear visual indication that the actuator had reached the target position. This stop-and-illuminate state was maintained for 5 seconds before the system re-evaluated the button state.

The mid-cycle interrupt check (if (digitalRead(buttonPin) == LOW) return;) inserted between the extend and retract phases ensured that a button press during the extension phase would prevent the retraction phase from executing, avoiding unnecessary actuator movement.

The table below summarises the observed system states:

Button State	Relay 1 (D7)	Relay 2 (D8)	Actuator Behaviour	LED (D2)
Not pressed	HIGH	LOW	Extending (5 s)	OFF

Not pressed	LOW	HIGH	Retracting (5 s)	OFF
Pressed (anytime)	LOW	LOW	Stopped immediately	ON
Pressed (between phases)	LOW	LOW	Retraction skipped	ON

The internal pull-up resistor configured on pin D3 (INPUT_PULLUP) ensured a stable HIGH reading when the button was open, eliminating the need for an external resistor and preventing floating-pin false triggers. No relay chatter or LED flickering was observed during testing, confirming stable digital logic transitions throughout the cycle.

8.0 DISCUSSION

The experiment successfully demonstrated the control of a 12V DC linear actuator using two different control methods: the L298P motor driver shield and the 2-way channel relay module. Both methods were able to control the actuator direction effectively; however, their operating principles and performance characteristics differed significantly.

From the experiment using the L298P motor driver shield, it was observed that Pulse Width Modulation (PWM) played an important role in controlling actuator speed. By varying the PWM values, the average voltage supplied to the actuator changed, resulting in different actuator speeds. Higher PWM values produced faster actuator movement, while lower PWM values reduced the movement speed. This showed that PWM is an efficient method for speed control because it allows smooth motor operation without changing the actual supply voltage.

The direction control using the L298P motor driver was also effective. Changing the logic state of the direction pin reversed the polarity supplied to the actuator, causing it to extend or retract. The actuator movement was smooth and stable throughout the experiment, indicating that the motor driver shield provided reliable motor control performance.

In comparison, the relay module experiment demonstrated a simpler method for directional control. The relay successfully reversed the polarity supplied to the actuator by switching between the Normally Open (NO) and Normally Closed (NC) contacts. This allowed the actuator to extend and retract correctly. However, unlike the L298P motor driver, the relay module could not provide speed control because relays function only as mechanical ON/OFF switches. As a result, the actuator always operated at full speed whenever power was supplied.

The experiment also highlighted the importance of proper circuit protection and safety precautions. Removing the OPT jumper from the L298P shield prevented possible damage to the Arduino caused by back-EMF generated from the actuator. In addition, verifying wiring polarity and maintaining secure electrical connections contributed to stable actuator operation and prevented short circuits.

Several limitations were observed during the experiment. Minor delays occurred during relay switching due to the mechanical operation of the relay contacts. Furthermore, the actuator speed estimation was based mainly on visual observation rather than precise measurement instruments, which may reduce measurement accuracy. Future improvements could include using sensors such as encoders or limit switches to obtain more accurate motion analysis and automatic position control.

Overall, the experiment provided a better understanding of motor driver operation, PWM speed control, relay switching mechanisms, and Arduino-based actuator control systems. The objectives of the experiment were achieved successfully, and the system operated according to the expected behavior.

9.0 CONCLUSIONS

In conclusion, the experiment successfully demonstrated the control of a 12V DC linear actuator using two different methods: the L298P Motor Driver Shield and the 2-channel relay module. Both systems were capable of controlling the actuator’s extension and retraction by reversing the polarity supplied to the actuator terminals. However, each method exhibited different operational characteristics and advantages. The L298P motor driver shield provided both speed and direction control through the implementation of Pulse Width Modulation (PWM) and digital direction signals. The experiment confirmed that increasing the PWM duty cycle increased the actuator speed, resulting in smoother and more flexible motion control. In contrast, the relay module offered a simpler approach for directional control by using electromechanical switching to reverse actuator polarity. Although the relay circuit could not regulate speed, it provided reliable full-speed operation with good electrical isolation between the control and power circuits. The modified relay system integrated with a push-button and LED indicator also demonstrated effective position detection and automatic actuator stopping functionality. Overall, the objectives of the experiment were achieved successfully, and the laboratory session enhanced understanding of motor drivers, relay switching mechanisms, PWM control, actuator operation, and Arduino-based embedded control systems.

10.0 RECOMMENDATIONS

Several improvements can be implemented in future experiments to enhance the accuracy, safety, and functionality of the actuator control system. Firstly, sensors such as limit switches, encoders, or position sensors can be integrated into the system to provide more precise feedback on actuator position and movement. This would enable automatic stopping, closed-loop control, and improved motion accuracy. In addition, measuring instruments such as tachometers or current sensors could be used to obtain more accurate quantitative data on actuator speed, current consumption, and system efficiency instead of relying solely on visual observation. For the relay-based system, replacing mechanical relays with solid-state relays or transistor-based H-bridge circuits could reduce switching delays, minimize mechanical wear, and improve overall response time. Proper heat sinks and cooling methods are also recommended for prolonged actuator operation to prevent overheating of the motor driver components. Furthermore, implementing safety protection features such as fuse protection, emergency stop switches, and flyback diodes would improve system reliability and reduce the risk of electrical damage caused by back-EMF or short circuits. Future work may also include developing wireless or IoT-based actuator control systems for remote monitoring and automation applications.

11.0 REFERENCES

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APPENDICES

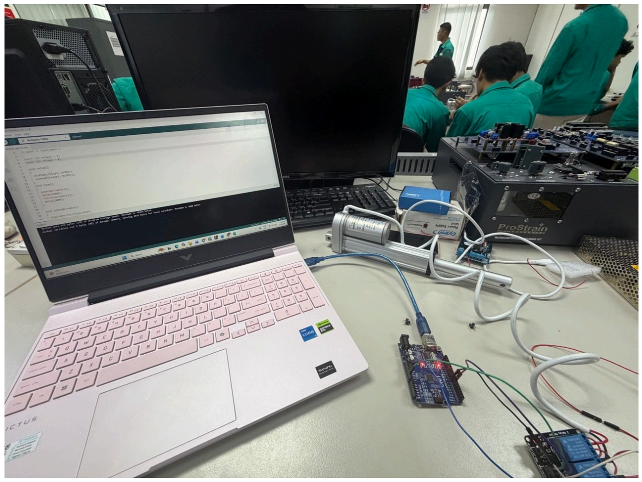

Figure 1.0: Uploading Code to the Arduino Uno

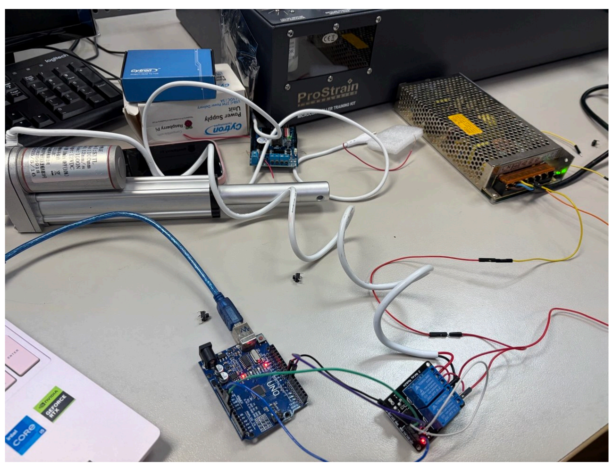

Figure 1.1: Setup of the experiment

ACKNOWLEDGEMENTS

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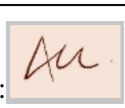
STUDENTS' DECLARATION

<u>Certificate of Originality and Authenticity</u>

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgment, and that the original work contained herein has not been untaken or done by unspecified sources or persons. We hereby certify that this report **has not been done by only one individual**, and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have **read** and **understand** the content of the total report, and no further improvement on the reports is needed from any of the individual contributors to the report. We therefore agreed unanimously that this report shall be submitted for **marking**, and this **final printed report** has been **verified by us**.

Signature:  Name: Amir Aiman Bin Ariff Affendi Matric Number: 2315925 Contributions: Introduction, Materials and Equipment, Experimental setup, Results	Read [/] Understand [/] Agree [/]
Signature:  Name: Anis Aqilah Binti Abdullah Matric Number: 2329678 Contributions: Abstract, Conclusion, Recommendations, References, Appendices, Acknowledgements	Read [/] Understand [/] Agree [/]
Signature:  Name: Azril Ulwan Bin Mohammad Imri Matric Number: 2314617 Contributions: Methodology, Data Collection, Data Analysis, Discussions	Read [/] Understand [/] Agree [/]